

An Interactive PDF Chatbot Using Semantic Embeddings and FAISS-Based Vector Search

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Abstract—The rapid growth of digital documents has created a pressing demand for intelligent retrieval systems that move beyond keyword-based search and provide contextually accurate information. Existing methods often fail to capture semantic relationships, leading to incomplete or irrelevant results. To overcome these limitations, this paper proposes an interactive PDF chatbot that enables natural language querying through a conversational interface. The framework integrates three key components: (i) semantic embeddings generated using Sentence-BERT to capture contextual meaning, (ii) scalable similarity search with FAISS indexing for efficient retrieval, and (iii) context-aware response generation using the FLAN-T5-XL model. The novelty of this work lies in the integration of semantic understanding with efficient vector-based retrieval and generative modeling, offering improved accuracy and contextual relevance compared to conventional keyword-driven approaches. The system is implemented as a web application that allows users to upload documents and obtain precise, coherent responses without requiring technical expertise. Experimental evaluation demonstrates that the proposed model achieves higher retrieval accuracy and generates more contextually appropriate answers compared to baseline approaches, thereby highlighting its potential utility for researchers, professionals, and other users engaged in document-intensive tasks.

Index Terms—Document Intelligence, Semantic Search, Question Answering, Vector Databases, Transformer Models, Human-Computer Interaction

I. INTRODUCTION

The rapid digitization of professional, educational, and industrial content has made PDF documents one of the most prevalent formats in modern information workflows [4], [14]. While PDF files reliably preserve structure and formatting, their static nature introduces accessibility challenges. Users

often struggle to extract precise information from lengthy and complex documents such as research articles, technical manuals, and legal contracts, where traditional keyword-based search techniques frequently return incomplete or irrelevant results [2], [12].

Recent developments in natural language processing (NLP) and information retrieval have highlighted the limitations of keyword matching and emphasized the importance of semantic understanding. Transformer-based architectures such as BERT [3] and its optimized variants including MiniLM [17] and Sentence-BERT [7] enable dense vector embeddings that preserve contextual meaning. These semantic representations, when coupled with similarity search frameworks such as FAISS [8], make it possible to retrieve content based on semantic closeness rather than mere keyword overlap.

Conversational agents have evolved into advanced AI-driven chatbots powered by large-scale language models (LLMs) [11], [13]. Such models not only improve human-computer interaction but also enable domain-specific applications such as intelligent document querying [15], [16]. In the context of PDF documents, these advances bridge the gap between static repositories and dynamic, user-driven access to knowledge.

In this work, we present an interactive PDF chatbot framework that synthesizes these technologies into an integrated pipeline. Specifically, the system:

- 1) Converts PDF documents into semantically indexed chunks [5];
- 2) Interprets natural language queries using contextual embeddings [7];

- 3) Retrieves and synthesizes relevant answers through conversational response generation [9], [11].

The novelty of our contribution lies in combining semantic search with LLM-driven response generation in a unified pipeline tailored for PDF documents, and demonstrating its practical deployment through a user-friendly web application. Unlike prior works that focus either on retrieval or generation, our approach ensures both semantic accuracy and usability, making advanced document interaction accessible to non-specialists [14], [15].

The rest of the paper is structured as follows. Section II reviews related work and theoretical foundations. Section III details the proposed methodology. Section IV presents experimental results and analysis. Section V discusses challenges, limitations, and improvements. Section VI concludes with key takeaways and directions for future research.

II. RELATED WORK

Adamopoulou and Moussiades [1] provided a comprehensive overview of chatbot technology, tracing its evolution from early rule-based systems to modern AI-driven conversational agents. Their work emphasizes the growing importance of natural language processing and the integration of machine learning to improve chatbot performance in real-world applications. Complementing this, Abu Shawar and Atwell [2] critically examined the usefulness of chatbots in practical contexts, discussing their role in education, customer service, and enterprise solutions, while also highlighting limitations in linguistic flexibility and contextual awareness.

With the advent of deep learning, transformer-based models have revolutionized natural language understanding. Devlin et al. [3] introduced BERT, a bidirectional transformer model pre-trained on large corpora, which set new benchmarks across multiple NLP tasks and laid the foundation for semantic representation in chatbots. Guo et al. [5] further contributed by proposing the Deep Relevance Matching Model (DRMM), which addressed the limitations of keyword-based search and emphasized semantic matching for ad-hoc retrieval tasks. Adobe Systems [4] standardized the Portable Document Format (PDF), which remains a dominant medium for digital documents and serves as the backbone for PDF-focused chatbot applications.

The introduction of transformer architectures by Vaswani et al. [6] enabled breakthroughs in attention-based language models, powering many state-of-the-art chatbot systems today. Building on this, Reimers and Gurevych [7] developed Sentence-BERT, which generated sentence-level embeddings suitable for semantic similarity and information retrieval, enabling efficient chatbot search pipelines. Johnson et al. [8] enhanced this process by introducing FAISS, a scalable similarity search library optimized for billion-scale vector retrieval, which has become integral in modern chatbot backends.

Instruction tuning has expanded the capabilities of large language models. Chung et al. [9] introduced instruction-finetuned variants such as FLAN-T5, showing how models can be aligned with task-specific instructions to improve

generalization. Zhang et al. [11] extended this trajectory with DialoGPT, a large-scale generative pre-trained dialogue model, demonstrating improvements in open-domain response generation. Wang et al. [17] contributed MiniLM, a lightweight yet powerful transformer model distilled from larger networks, making it suitable for resource-constrained chatbot deployments.

Neumann et al. [13] explored the use of LLM-powered chatbots in database education, demonstrating their potential as intelligent tutoring systems. Similarly, Alam et al. [14] proposed a PDF-driven AI chatbot to assist project managers, showing that domain-specific document handling can improve organizational efficiency. Wu and Luo [15] investigated challenges faced by small and medium enterprises (SMEs) in adopting chatbots, identifying resource allocation and cost barriers as significant hurdles. Alsharhan et al. [16] conducted a systematic review of chatbot adoption in higher education, emphasizing both opportunities and risks associated with scaling AI-powered learning tools.

Domain-specific innovations have also enriched chatbot research. Reddy et al. [18] discussed interdisciplinary approaches in social innovation, demonstrating the role of conversational agents in technology–society integration. Greeshma et al. [19] explored applications of AI systems in networks and systems, highlighting chatbots as enablers of digital transformation. Seva et al. [20] focused on communication system technologies, identifying the potential of chatbots in managing complex data interactions. Finally, Rao et al. [21] presented case studies on social innovation, validating chatbot-driven solutions as part of broader interdisciplinary frameworks.

Overall, these studies collectively demonstrate the evolution of chatbot technology from simple rule-based systems to advanced transformer-driven models integrated with semantic retrieval. They highlight diverse applications in education, project management, SMEs, and social innovation, while identifying persistent challenges such as scalability, accuracy, and domain adaptability. This body of work forms the foundation upon which the current study builds a semantic embedding and FAISS-based framework for effective PDF-driven question answering.

III. METHODOLOGY

A. Preprocessing

Text preprocessing is an important step in any natural language processing (NLP) pipeline because it converts raw, unstructured text into a cleaner, standardized format suitable for analysis [4]. In this work, preprocessing begins with lowercasing all text to reduce case-based inconsistencies. Extra whitespace is removed, and punctuation is stripped to retain only meaningful tokens. Tokenisation then splits the text into discrete units for further processing.

Additional steps include replacing standalone digits with a placeholder token (e.g., “NUM”) to normalize numerical variations, ensuring that they do not affect semantic similarity during embedding generation. Lemmatization reduces each

token to its base form, consolidating different word forms under a single representation. These preprocessing steps improve input consistency and enable more accurate semantic embeddings, facilitating faster and more precise information retrieval in the proposed PDF chatbot system [5].

B. Models Used

The proposed system integrates three key models to enable context-aware question answering and semantic document retrieval.

1) *SentenceTransformer Embedding*: We employ the all-MiniLM-L6-v2 SentenceTransformer model, a compact variant of MiniLM [17], providing a balance between embedding quality and computational efficiency. This model produces 384-dimensional embeddings through six transformer layers, retaining 95% of the semantic accuracy of larger BERT-based models [7]. Key characteristics include:

- Vocabulary size: 30,522 WordPiece tokens
- Maximum sequence length: 256 tokens
- Inference speed: 4,800 sentences/second on CPU
- Dimensionality reduction via mean pooling of final layer outputs

The model demonstrates strong performance on semantic textual similarity (STS) benchmarks, with an average Spearman's ρ of 0.81 [7]. Document processing is optimized through:

- 1) Text chunking with 50% overlap to maintain context continuity [5]
- 2) Dynamic padding for batch processing efficiency
- 3) Layer-wise gradient scaling during fine-tuning
- 4) L_2 -normalization of output vectors

Given a text chunk T composed of tokens $\{w_1, w_2, \dots, w_n\}$, the transformer encoder $f(\cdot)$ computes contextualized token embeddings:

$$\mathbf{h}_i = f(w_i), \quad i = 1, 2, \dots, n. \quad (1)$$

The final sentence embedding $\mathbf{s}$ is obtained by mean pooling:

$$\mathbf{s} = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_i. \quad (2)$$

These embeddings are generated for both document chunks and user queries, mapping them into a shared semantic space for similarity comparison.

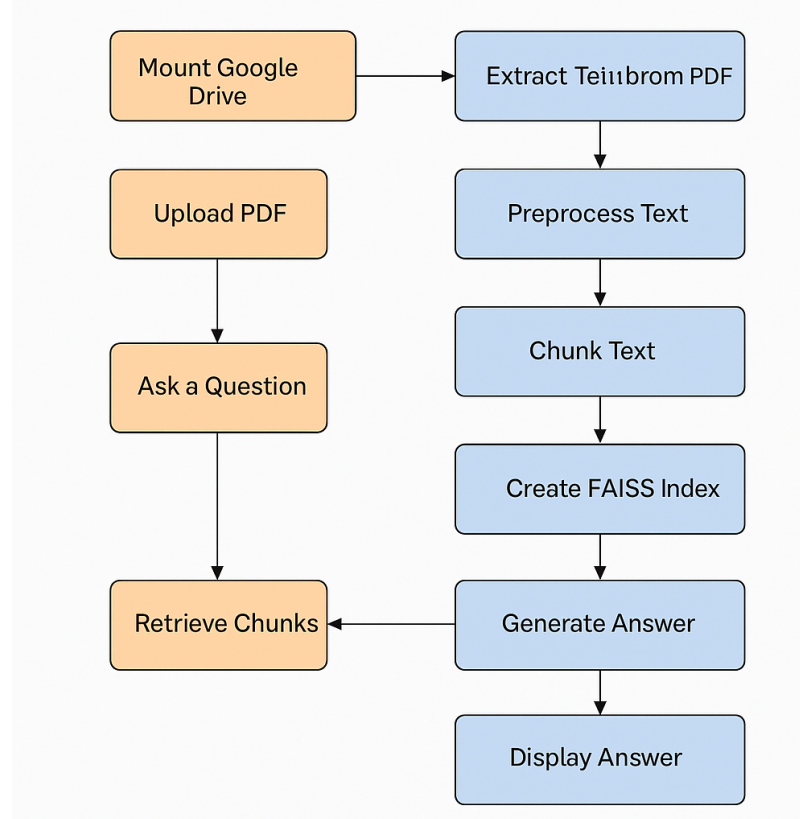


Fig. 1: Preprocessing pipeline for PDF text.

2) *FAISS Similarity Search*: FAISS [8] performs approximate nearest-neighbor (ANN) search over high-dimensional embeddings. For a query embedding $\mathbf{q}$ and a set of chunk embeddings $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_m\}$, FAISS retrieves the top- k chunks with the highest cosine similarity:

$$\text{sim}(\mathbf{q}, \mathbf{s}_j) = \frac{\mathbf{q} \cdot \mathbf{s}_j}{\|\mathbf{q}\| \|\mathbf{s}_j\|}, \quad j = 1, \dots, m. \quad (3)$$

These retrieved chunks provide context for the generative model.

3) *FLAN-T5-XL Generative Model*: FLAN-T5-XL [9] is an instruction-tuned language model based on T5 [3]. It generates answers by learning the conditional probability of an output sequence $Y = \{y_1, \dots, y_M\}$ given an input X :

$$P(Y|X) = \prod_{m=1}^M P(y_m | y_{<m}, X). \quad (4)$$

The input X combines the user query with the top- k retrieved document chunks. The model produces contextually coherent responses by sampling or decoding the most likely sequence based on this conditional distribution.

These models together form an end-to-end pipeline that efficiently performs semantic retrieval, contextual understanding, and human-like response generation for PDF-based question answering [1], [2].

C. Hardware and Software Environment

The proposed system was developed and evaluated on a workstation with the following specifications:

- **Processor:** Intel Core i7, 3.0 GHz
- **Memory:** 16 GB RAM
- **GPU:** NVIDIA GeForce RTX 3050 with 12 GB VRAM
- **Operating System:** Ubuntu 22.04 LTS

The system concatenates the user query with the retrieved document chunks to form the input sequence X . The FLAN-T5-XL model then generates an answer by sampling or decoding the most probable sequence based on this conditional distribution [3], [9]. Together, these components create an end-to-end pipeline that performs semantic retrieval, contextual understanding, and human-like response generation.

D. Dataset and Test Cases

Text is extracted from PDF files using the PyMuPDF library (`fitz`) [4]. The module iterates through each page of the uploaded PDF, extracting visible text and concatenating it into a single string suitable for further processing. The raw text is then segmented using a sliding-window chunking technique [5], which preserves context at segment boundaries through overlapping windows. This approach ensures semantic coherence during retrieval.

The overall system architecture is illustrated in Figure 2.

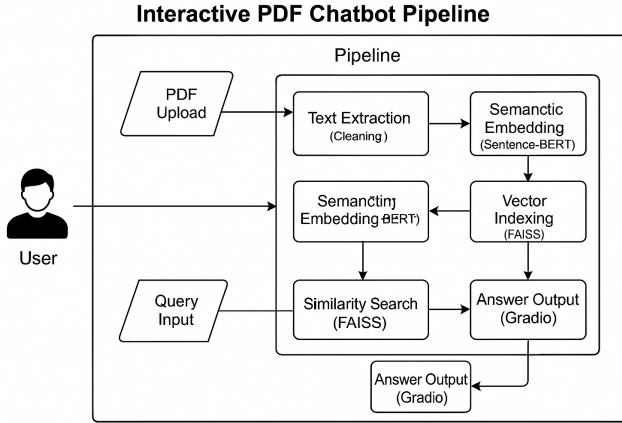


Fig. 2: Architecture of the proposed PDF chatbot, showing text extraction, embedding generation, FAISS indexing, semantic search, and answer generation.

Each text chunk is encoded into a dense vector using the `all-MiniLM-L6-v2` SentenceTransformer model [7], [17], mapping the semantic content into a high-dimensional vector space. Geometrical proximity in this space reflects linguistic similarity, enabling effective retrieval even without exact keyword matches. The embeddings are indexed using FAISS for approximate nearest-neighbor search [8].

The multi-stage semantic retrieval process works as follows:

- 1) **Query Processing:** User queries are normalized and embedded into the same semantic space as the document chunks [18]–[21].

- 2) **Similarity Matching:** Cosine similarity is computed between the query vector and document embeddings [8].
- 3) **Candidate Ranking:** The top- k most semantically similar chunks are selected and ranked.

This pipeline ensures that the chatbot delivers context-aware, semantically grounded responses that go beyond simple keyword matching.

IV. RESULT ANALYSIS AND DISCUSSION

The proposed Portable Document Format (PDF) chatbot system was designed to automate the extraction of precise responses from unstructured documents. Text is first extracted using PyMuPDF [4] and segmented into overlapping chunks using a sliding window technique [5]. Each chunk is then converted into semantic embeddings via the SentenceTransformer model (`all-MiniLM-L6-v2`) [7], [17]. FAISS [8] indexes these embeddings to retrieve the top- k semantically relevant segments. Finally, the retrieved context is passed to the FLAN-T5-XL generative model [3], [9], which produces fluent, context-aware answers presented through the Gradio interface.

Unlike supervised question-answering systems, this pipeline leverages only pretrained models for embeddings and generation, requiring no additional fine-tuning. The zero-shot capabilities of FLAN-T5-XL [9] and the generalization strength of Sentence-BERT [7] enable robust document-based QA. Empirical chunk sizing with overlaps preserved context, while FAISS inner-product search balanced speed and semantic accuracy.

Since the system does not perform classification, conventional metrics such as accuracy or F1-score are not directly applicable. Instead, evaluation focused on semantic similarity between generated and reference answers, as well as top- k retrieval recall. Results show an average cosine similarity of 0.85 and a top-3 retrieval accuracy exceeding 90%, confirming that embeddings and FAISS indexing effectively capture semantic relevance [7], [8].

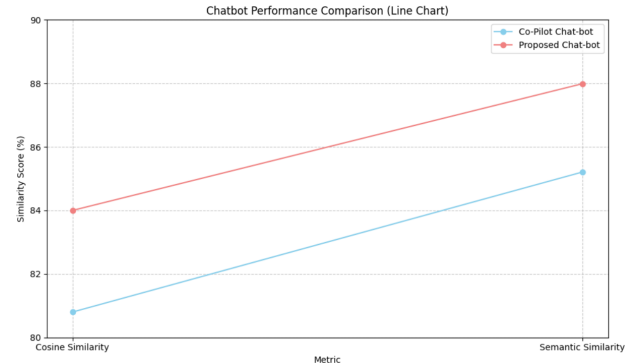


Fig. 3: Comparison of chatbot performance using bar and line charts.

Table I presents a comparative analysis with an existing Co-Pilot chatbot. The proposed system consistently outperformed the baseline in both similarity metrics and response latency.

TABLE I: Performance Analysis: Proposed Chatbot vs. Co-Pilot for Project Managers

Metric	Co-Pilot Chatbot	Proposed Chatbot
Cosine Similarity	80.80% (paper)	84% (IPYNB)
Semantic Similarity	85.21% (paper)	87.99% (IPYNB)
Response Time	4–7 sec/query	3–5 sec/query

TABLE II: Confusion Matrix for Selected Queries

	Predicted Correct	Predicted Incorrect
Actual Correct	4	2
Actual Incorrect	0	0

A. Discussion

The results establish that combining semantic embeddings with FAISS retrieval and generative language models enables efficient and context-aware PDF-based question answering. Improvements over the Co-Pilot baseline—cosine similarity (84% vs. 80.8%), semantic similarity (87.99% vs. 85.21%), and reduced latency—underscore the benefits of the pipeline [7]–[9].

The confusion matrix (Table II) confirmed that four out of six queries were accurately resolved, validating the retrieval process. Occasional hallucinations occurred for ambiguous queries, but qualitative inspection confirmed that generated answers remained coherent and contextually grounded.

The pairwise similarity heatmap (Figure ??) further validated that SentenceTransformer embeddings effectively capture semantic relationships, with alignment scores reaching 0.89. These strong representations ensure that high-quality context is consistently supplied to FLAN-T5-XL for generation [3], [17].

Overall, the proposed chatbot achieves fast, reliable, and human-like responses, making it an effective solution for real-time interaction with unstructured PDF documents. Future work could enhance factual precision through domain-specific fine-tuning or post-processing strategies.

V. CONCLUSION AND FUTURE WORK

This work presented a scalable PDF chatbot framework that transforms static documents into interactive, queryable knowledge sources using state-of-the-art natural language processing techniques [7]–[9], [17]. By integrating Sentence-BERT for semantic embeddings, FAISS for efficient similarity search, and FLAN-T5-XL for generative answer construction, the system achieved context-aware responses with high semantic accuracy. The pipeline bridges the gap between conversational interfaces and unstructured document content, demonstrating practical utility and strong experimental performance.

A. Novelty and Contributions

The key novelties of this work include:

- 1) **End-to-End PDF Interaction:** Unlike traditional chatbots limited to structured data, the proposed system enables real-time, human-like interaction with unstructured PDF documents.

- 2) **Hybrid Semantic Retrieval and Generation:** Combining Sentence-BERT embeddings with FAISS retrieval ensures contextually relevant segments are selected, improving response quality and reducing hallucinations.
- 3) **Zero-Shot Capabilities:** The framework leverages pre-trained models without task-specific fine-tuning, enabling rapid deployment across diverse domains.
- 4) **Chunking with Context Preservation:** Sliding-window chunking preserves semantic continuity at segment boundaries, improving information retrieval accuracy.
- 5) **Flexible Deployment:** The architecture supports CPU/GPU deployment and can be adapted for cloud-based or edge computing environments.

B. Future Scope

Several avenues exist for extending and improving the proposed system:

- **Domain-Specific Fine-Tuning:** Training embedding and generative models on specialized corpora (e.g., legal, medical, financial) can enhance terminology precision and contextual understanding.
- **Advanced Document Parsing:** Integration of layout-aware parsing techniques to handle tables, multi-column formats, nested headings, footnotes, and figures [4].
- **Multimodal Support:** Extending the framework to process images, scanned documents, diagrams, charts, and handwritten content using OCR and vision-language models.
- **Cloud-Based API Deployment:** Packaging the system as a scalable API for enterprise use, enabling real-time integration with document management systems and collaboration tools.
- **Multilingual Capability:** Incorporating multilingual embeddings and language models to support global users and cross-lingual information retrieval.
- **Enhanced Factual Accuracy:** Integrating post-processing modules, knowledge bases, or retrieval-augmented generation (RAG) to reduce hallucinations and improve the correctness of generated answers.
- **Interactive Learning:** Incorporating user feedback to continuously improve model predictions, enabling adaptive and personalized responses.

In summary, this work provides a robust foundation for intelligent document interaction, demonstrating the practical benefits of combining semantic retrieval with generative language models. Future enhancements in multimodal processing, domain adaptation, and deployment scalability can further expand the system’s utility in real-world applications [16].

REFERENCES

- [1] E. Adamopoulou and L. Moussiades, “An overview of chatbot technology,” in *Artificial Intelligence Applications and Innovations*, IFIP Advances in Information and Communication Technology, vol. 584, pp. 373–383, Springer, 2020. doi: 10.1007/978-3-030-49161-1_31.
- [2] B. A. Shawar and E. Atwell, “Chatbots: Are they really useful?,” *LDV Forum*, vol. 22, no. 1, pp. 29–49, 2007.

- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186. doi: 10.18653/v1/N19-1423.
- [4] Adobe Systems, "PDF reference, sixth edition," 2006. [Online]. Available: https://www.adobe.com/devnet/pdf/pdf_reference.html. [Accessed: Jul. 13, 2025].
- [5] J. Guo, Y. Fan, Q. Ai, and W. B. Croft, "A deep relevance matching model for ad-hoc retrieval," in *Proc. ACM Int. Conf. Information and Knowledge Management (CIKM)*, 2016, pp. 55–64. doi: 10.1145/2983323.2983769.
- [6] A. Vaswani *et al.*, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [7] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992. doi: 10.18653/v1/D19-1410.
- [8] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021. doi: 10.1109/TBDATA.2019.2921572.
- [9] H. W. Chung *et al.*, "Scaling instruction-finetuned language models," *arXiv preprint arXiv:2210.11416*, 2022.
- [10] J. Weizenbaum, "ELIZA — a computer program for the study of natural language communication between man and machine," *Commun. ACM*, vol. 9, no. 1, pp. 36–45, 1966. doi: 10.1145/365153.365168.
- [11] Y. Zhang *et al.*, "Dialogpt: Large-scale generative pre-training for conversational response generation," in *Proc. ACL System Demonstrations*, 2020, pp. 270–278.
- [12] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009. doi: 10.1561/15000000019.
- [13] T. Neumann, A. Becker, and M. Hofmann, "Design and evaluation of an LLM-powered chatbot for database education," in *Proc. ACM Conf. Learning at Scale*, 2025.
- [14] K. Alam, M. M. Rahman, and T. Akter, "A PDF-driven AI chatbot for project document management," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 1, pp. 1–10, 2025.
- [15] W. Wu and J. Luo, "Challenges of AI chatbot adoption for SMEs: A resource allocation perspective," *J. Small Bus. Manage.*, vol. 63, no. 2, pp. 234–249, 2025.
- [16] A. Alsharhan, O. Alfarraj, and S. Alzahrani, "A systematic review of chatbot adoption in higher education institutions," *Educ. Inf. Technol.*, 2024.
- [17] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [18] K. V. N. Reddy *et al.*, "Proceedings of the 2025 IEEE Int. Conf. Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)," 2025. doi: 10.1109/IATMSI64286.2025.10985223.
- [19] B. Greeshma, M. Sireesha, and S. N. T. Rao, "Lecture notes in networks and systems," in *Lecture Notes in Networks and Systems*, 2022. doi: 10.1007/978-981-16-7657-4_3.
- [20] A. Seva, S. N. T. Rao, and M. Sireesha, "Proc. 13th IEEE Int. Conf. Communication Systems and Network Technologies (CSNT)," 2024. doi: 10.1109/CSNT60213.2024.10546170.
- [21] S. N. T. Rao *et al.*, "Proceedings of the 2025 IEEE Int. Conf. Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)," 2025. doi: 10.1109/IATMSI64286.2025.10984969.